Units 3 and 4

Practice exam suggested answers

Section A—Multiple-choice questions

1. B. The reaction between propene and bromine. This reaction is an addition reaction that will have one product only; therefore, the atom economy is 100%.

2. B. 3- Methylhexan-2-ol. The chain contains six carbon atoms, so the name of the molecule will be hexane based. Numbering for this molecule should start from the right-hand side. The second carbon has an alcohol group and the third carbon has a methyl group; hence, the systematic name is 3-methylhexan-2-ol.

3. D. +1942. The activation energy is always positive. The value for the reverse reaction is: 1370 + 572 = 1942 kJ mol^{-1}

4. D. $Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^-$. Pt is an unreactive metal used as the electrode. There is no Fe(s) in the cell. C is not correct because the charge on the reactant side is not equal to the charge on the product side (3+ : 1+).

5. D. Binding to the enzyme active site to block another compound from doing so. Lock-and-key enzymes require the substrate to fit into the active site exactly. With the competitive enzyme inhibitor in the active site already, the enzyme is no longer able to work. A: The active site is only changed by changes to the tertiary structure caused by temperature or pH changes. B: An alternative active site is not created; but even if it were it would need to be identical, which is unlikely. C: Denaturing involves the changing of the tertiary bonds by pH or temperature.

6. C. Butanoic acid. In this process, butane may form chlorobutane. Chlorobutane will form butan-1-ol and this will be converted to butanoic acid.

7. A. $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$. The graph shows two reactants and one product. One reactant drops at half the rate of the other, so the ratio of reactants is 2 : 1, which matches with option A. B: This reaction is a 1 : 1 ratio, so all changes would be the same size on the graph. D and C: Each option has only has one reactant, which does not match the graph.

8. A. 1.9 mol. The amount of NOCl drops by 0.6 mol. Therefore, the amount of NO should increase by 0.6 mol and the amount of Cl_2 will increase by half as much, 0.3 mol. The total number of moles will therefore be 1.0 + 0.6 + 0.3 = 1.9 mol.

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9. C. Ethyl ethanoate. The structure of ethyl ethanoate is $CH_3CH_2OCOCH_3$. This molecule will have three different hydrogen environments with a splitting pattern matching the spectrum provided.
A: Propanol would have 3 environments, a quartet (CH_2), a triplet (CH_3) and a singlet (OH).
B: Methyl ethanoate would have 2 environments of singlets as neither CH_3 has any adjacent protons.
D: Butanoic acid will have four environments.

10. D. Might share a similar parent molecular ion but differ in most other peaks. Propane and carbon dioxide both have a relative molecular mass of 44, so they share a parent molecular ion of *m*/*z* 44. Most other peaks will be different, however, due to different splitting patterns.

11. B. 0.170 g

$q = 4.18 \times 120 \times 10.0 = 5016 = 5020$ J

$n(\text{ethanol}) = \dfrac{5.016}{1360} = 0.003\,69$ mol

Mass of ethanol = $0.003\,69 \times 46.0 = 0.170$ g

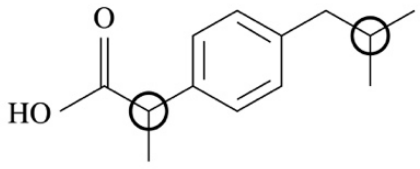
12. B. The breaking of bonds in nitrogen and hydrogen is endothermic and the formation of N–H bonds exothermic. The breaking of reactant bonds is endothermic and the formation of product bonds is exothermic.

13. D. +5. VO_3^{-1}; $x + (-2 \times 3) = -1$, so $x = +5$. NH_4 is present as a +1 polyatomic ion. Charge of –1 on VO_3 ion is not explicit.

14. B. Oxidation occurs at the anode and anions move into the cell. All other answers use a combination of oxidation or reduction, and movement of anions or cations.

15. B. Hydrolysis. Water is required to the break the ester bond in lipids and the glycosidic link in the carbohydrates. A: Condensation is when water is a product of the reaction, this occurs when lipids and carbohydrates form. C and D: Addition and oxidation are not relevant.

16. A. Ibuprofen. Chiral centres occur when there are four different substituents on the carbon. This rules out any carbons with a double bond and carbons with 2 hydrogens attached. Chiral centres are shown circled here.



ibuprofen

17. D. Tertiary. Bonds in the tertiary structure are the protein structure that is the overall three-dimensional shape of the entire protein molecule. A: A peptide link forms inside the primary structure. B: The primary structure is the sequence of amino acids. C: The secondary structure is the coiling and pleating caused by H-bonds.

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18. A. Transesterification. B: Combustion is the burning of fuels in oxygen. C: Fermentation is the production of ethanol from glucose. D: Polymerisation is the formation of polymers.

19. D. I and III only. When reversed, the top reaction is oxidation, which will happen at the anode, which is positive; the bottom reaction will be reduction, which occurs at the negative cathode. Thus Cd(s) is produced at the negative electrode. II is not correct because the $2OH^-$ produced at the cathode is cancelled out by the $2OH^-$ consumed at the anode, so there will be no change in pH. III is always correct regardless of discharge or recharge.

20. B. 2-chlorohexane. This will be the most volatile because the dipole–dipole intermolecular bonds formed by the Cl atoms will not be as strong as the dipole–dipole bonds formed by the higher electronegativity of the oxygen in the other structures. Volatility relates to the temperature with which the liquid turns into gas.

21. B. Electrolytic cells. These are the only cells that cannot be spontaneous. Secondary galvanic cells can be non-spontaneous when they recharge, but they will be spontaneous when discharging.

22. D. Two amino groups, one at the terminus and one on the Lys R-group. Two carboxy groups, one on the terminus and one on the Asp R-group. Four amide groups due to the peptide links formed between each amino acid in the chain.

23. C. Fossil fuels are being used up faster than they can be formed. Renewability is related to the rate of production and use. It is renewable if it can be produced as fast or faster than it is used. Sustainability is about whether the resource can be used long term without harmful effects.

24. D. Using renewable energy sources including wind. Green hydrogen must be produced using renewable energy sources. All other answers also produce other by-product gases that are harmful to the environment.

25. B. Increase the rate of ethanol production. C: Ethanol has the highest boiling point of the three gases and it is lower than 100°C. Maintaining gaseous states would not require 300°C. A and D are incorrect because a high temperature will favour the back reaction as the equation is exothermic.

26. A. Phosphoric acid is a catalyst and does not take part in the reaction. B: Thermochemical equations must include the acid if it contributes to the energy of the system by forming or breaking bonds. C: Oxidising agents are required to convert ethanol to ethanoic acid, not acids.

27. A. Distillation. The process of heating to separate organic compounds by their boiling points is distillation. B: Volumetric analysis determines unknown concentrations, C: Vaporisation, while separating by boiling point, would not include the condensation step, which is how different compounds are collected. D. Chromatography does not require the use of a condenser and only gas chromatography uses high temperatures to vaporise components.

28. D. Amide. Dopa contains carboxyl, amine and hydroxyl functional groups. There is no CONH amide group.

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29. A. Carboxyl. The spectrum shows a broad peak at 3000 cm^{-1}, which is consistent with O–H (acid) and a narrow peak at approximately 1700 cm^{-1}, which is consistent with C=O. The O–H (alcohol) is not shown, as it would be a broad, smooth peak at 3200 cm^{-1}. There are also no N–H peaks, which would be shown as 'fangs', which look like two peaks in one between 3300 and 3500 cm^{-1}.

30. C. P_4O_6 by 2.78 g.

$n(P_4O_6) = \frac{m}{M} = \frac{4.40}{220.0}$ = 0.0200 mol

$n(I_2) = \frac{m}{M} = \frac{3.00}{253.8}$ = 0.011s

$n(P_4O_6) \times 5$ = 0.100 mol (excess reactant)

$n(I_2) \times 8$ = 0.0944 mol (limiting reactant)

Therefore $n(P_4O_6)$ needed = $0.0118 \times \frac{5}{8}$ = 0.007 38 mol, $m(P_4O_6)$ reacted = 0.007 38 × 220.0 = 1.62 g.

4.40 – 1.62 = 2.78 g

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Section B—Short-answer questions

Question 1a.

$C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$; $\Delta H = -1560$ kJ mol^{-1}

OR

$2C_2H_6(g) + 7O_2(g) \rightarrow 4CO_2(g) + 6H_2O(l)$; $\Delta H = -3120$ kJ mol^{-1} 2 marks

(1 mark for correctly balanced equation and correct states (water must be liquid as it is at SLC), 1 mark for correct ΔH (including sign and unit)).

Question 1b.

Heat of combustion of ethane = 51.9 kJ g^{-1}, so 5.00 × 51.9 = 260 kJ 1 mark

Question 1c.

1 mol C_2H_6 (g) produces 1560 kJ, therefore 1560 × 4.0 = 6240 kJ would be required to form 4.0 mol of ethane. 1 mark

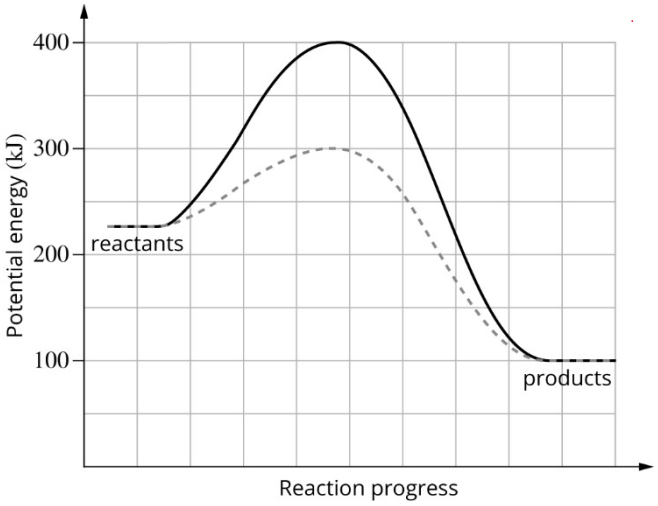
Question 1d.

Actual amount of energy required = $\frac{4.97 \times 10^{14}}{0.53}$ = 9.4 × 10^{14} MJ 1 mark

Ethane produces 51.9 kJ g^{-1}, which is equal to 51.9 MJ kg^{-1}, so $\frac{9.4 \times 10^{14}}{51.9}$ = 1.8 × 10^{13} kg. 1 mark

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Question 1e.



1 mark

Question 2a.

$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$

	2NO	**O_2**	**$2NO_2$**
initial	1.6	1.6	0
equilibrium	1.12	$1.6 - 0.48 \times \frac{1}{2}$	0.48

1 mark ($n(O_2)_e$ and $n(NO)_e$)

$$K = \frac{[NO_2]^2}{[NO]^2[O_2]} = \frac{0.48^2}{(1.12)^2 \times 1.36}$$

1 mark

$= 0.14\ M^{-1}$

1 mark (unit must be included for this mark)

Question 2b.

	Impact upon *K*	**Impact on the amount of CO**	**Impact on the concentration of CO**
Decrease in temperature	Decrease in *K*	Decrease	Decrease
Decrease in volume	No change	Decrease	Increase

6 marks

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Question 2ci.

$K = \frac{1}{12.4} = 0.0806$, $\Delta H = +132$ kJ mol^{-1} 1 mark

Question 2cii.

$K = \frac{1}{\sqrt{12.4}} = 0.290$, $\Delta H = +66$ kJ mol^{-1} 1 mark

Question 2d.

K for this equation = $\sqrt{12.4}$; 1 mark

therefore, the equation is $H_2(g) + NO(g) \rightleftharpoons H_2O(g) + \frac{1}{2}N_2(g)$ 1 mark

Question 3a.

$CF = \frac{VIt}{\Delta T}$

$E = 8.20 \times 0.750 \times (10.0 \times 60) = 3.69$ kJ 1 mark

$CF = \frac{3.69}{5.88} = 0.627$ kJ °C^{-1} 1 mark

Question 3b.

$n(\text{Li}) = \frac{0.238}{6.9} = 0.034$ mol 1 mark

$E = CF \times \Delta T = 0.627 \times (32.38 - 18.45) = 8.67$ kJ 1 mark

$\Delta H = \frac{E}{\text{mol}} = -255$ kJ for reaction

2 mol Li(s) in equation so, ΔH for equation = $255 \times 2 = -510$ kJ mol^{-1} 1 mark

Question 3c.

Lithium is a critically endangered element in that it only exists in small quantities on Earth. This means that unless recycling is dramatically improved, its use is unsustainable. 1 mark

Lithium is very reactive with water and oxygen, making it quite dangerous. So replacing it with a safer chemical would be preferred. 1 mark

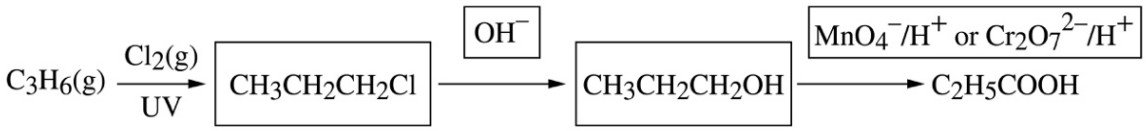
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Question 4a.

Test	Experiment conducted	Likely result	Chemical equation
1	Add bromine to each liquid. (0.5 mark)	Bromine will react with hexene and become colourless. Bromine will not react with propanoic acid. (0.5 mark)	$C_6H_{12}(l) + Br_2(l) \rightarrow C_6H_{12}Br_2(l)$ (1 mark)
2	Add a base and indicator to each liquid. (0.5 mark)	Propanoic acid will neutralise the base. Hexene will not react. (0.5 mark)	$C_2H_5COOH(aq) + NaOH(aq) \rightarrow C_2H_5COONa(aq) + H_2O(l)$ (1 mark)

4 marks

Question 4b.



4 marks (1 mark per box)

Question 5ai.

C_2H_4O; the molecular ion peak is 44, so to add up to this molar mass requires 2 × C, 4 × H and 1 × 16.

1 mark

Question 5aii.

15: CH_3^+ 1 mark

29: CHO^+ 1 mark

Question 5bi.

^1H NMR: two different hydrogen environments, one quartet, the other a doublet. 1 mark

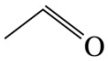
Question 5bii.

^{13}C NMR: two different carbon environments, peak at 200 ppm is RHC=O. 1 mark

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Question 5c.

Ethanal 1 mark



1 mark

Question 6a.

As the rhubarb is cut into smaller pieces, its surface area increases. As the surface area increases, the rate of reaction increases, making the times recorded smaller. 1 mark

Question 6b.

As the temperature increases, the average kinetic energy increases. Therefore, there will be more frequent collisions and a higher percentage of collisions with $E \geq E_a$, which produces an increase in the rate of reaction. 2 marks

Question 6c.

Time taken for the solution to become colourless. 1 mark

Question 6d.

Introduce multiple trials to test repeatability. 1 mark

Record a video of the experiment to reduce the uncertainty of the results introduced with subjective measurement of the time the solution becomes colourless. 1 mark

Question 7a.

Propane is non-polar, so the only intermolecular forces are dispersion forces. Propane will have the lowest boiling point. 1 mark

Methoxymethane will have similar dispersion forces and dipole–dipole forces acting. 1 mark

Ethanol will have similar dispersion forces as both other compounds, but is the only molecule to contain hydrogen bonds, which are the strongest intermolecular forces, so ethanol will have the highest boiling point. 1 mark

Question 7b.

Propane 1 mark

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Question 7c.

The compound that is most strongly adsorbed onto the stationary phase will have the highest retention time. The compounds that are most soluble in the mobile phase (the solvent) will have the shortest retention time. 1 mark

The size of a molecule will also impact its ability to move through the stationary phase, so larger molecules will have longer retention times. 1 mark

Question 7d.

Ethanol is the most polar. It will be the most soluble into the mobile phase, giving it the shortest retention time in the polar mobile phase. 1 mark

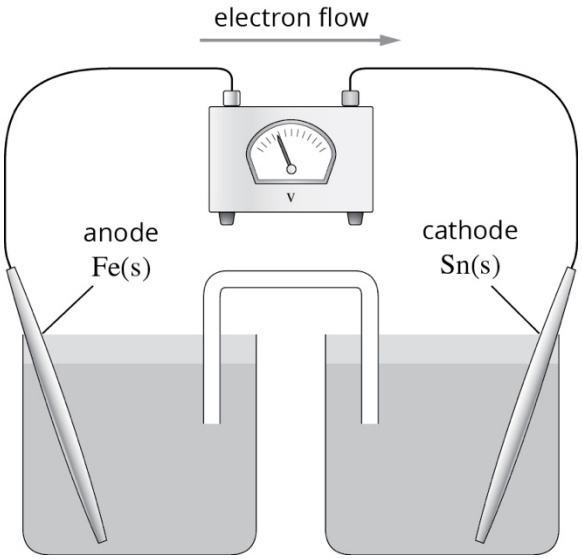
Question 7e.

The sample has low purity because, although it contains propane, it also contains small amounts of methane and acetylene and an unknown substance with a peak at 2.2 minutes that is not shown on the standard chromatogram. 1 mark

Question 8a.

$Fe(s) + Sn^{2+}(aq) \rightarrow Fe^{2+}(aq) + Sn(s)$ 1 mark

Question 8b.



Anode: Fe(s) 1 mark

Cathode: Sn(s) 1 mark

Electron flow from Fe(s) through external conductor to Sn(s). 1 mark

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Question 8c.

The salt bridge is the internal circuit containing ions that are free to move so that they can balance charges formed in the two compartments. 1 mark

The cations move to the cathode to replace cations being reduced and anions move to the anode to balance the cations being produced through oxidation. 1 mark

Question 8d.

–0.14 – (–0.44) = 0.30 V 1 mark

Question 8e.

Any two of:

- Half-cells need to be prepared at standard conditions (1 mol L^{-1} solutions, 25°C, 1 atm pressure).
- Potential decreases as reaction proceeds due to build-up of products, reduction of reactants.
- No information about the rate of the reaction. 2 marks

Question 9a.

Oxidation: $Be \rightarrow Be^{2+}(aq) + 2e^-$

Reduction: $MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l)$ 2 marks

(1 mark for each correctly balanced half equation. 1 mark for correct placement of oxidation and reduction reactions. States of matter not required.)

Question 9b.

$5Be(s) + 2MnO_4^-(aq) + 16H^+(aq) \rightarrow 5Be^{2+}(aq) + 2Mn^{2+}(aq) + 8H_2O(l)$ 2 marks

(1 mark for correct equation. 1 mark for correct states)

Question 10a.

Anode (positive): $O_2(g) + 4H^+(aq)$ 1 mark

Cathode (negative): Cu(s) 1 mark

Question 10b.

$n(Cu) = \frac{m}{M} = \frac{10.0}{63.5} = 0.157$ mol 1 mark

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$n(e^-) = n(Cu) \times \frac{2}{1} = 0.315$ mol 1 mark

$Q = n(e^-) \times F = 0.315 \times 96\,500 = 30\,394$ C 1 mark

$t = \frac{Q}{I} = \frac{30394}{2.40} = 12\,664$ s = 211 min (–1, if not 3 significant figures) 1 mark

Question 10c.

$Q = \frac{30394}{0.80} = 37\,993$ C 1 mark

$t = \frac{Q}{I} = \frac{37993}{2.40} = 15\,830$ s = 264 min

Extra time = 264 – 211 = 53 min 1 mark

Question 11a.

The carbon used as CO_2 to produce glucose during photosynthesis is converted to ethanol and carbon dioxide during fermentation. The ethanol can then be used in combustion, in which the carbon is released as CO_2. The total amount of CO_2 released during fermentation and combustion is equal to the amount of CO_2 used in photosynthesis. 1 mark for explanation

$n(CO_2$ in photosynthesis) = $n(CO_2$ in fermentation) + $n(CO_2$ in combustion)

Photosynthesis: $6CO_2(g) + 6H_2O(l) \rightarrow C_6H_{12}O_6(aq) + 6O_2(g)$ 1 mark

Fermentation: $C_6H_{12}O_6(aq) \rightarrow 2C_2H_5OH(aq) + 2CO_2(g)$ 1 mark

Question 11b.

$2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$ 1 mark

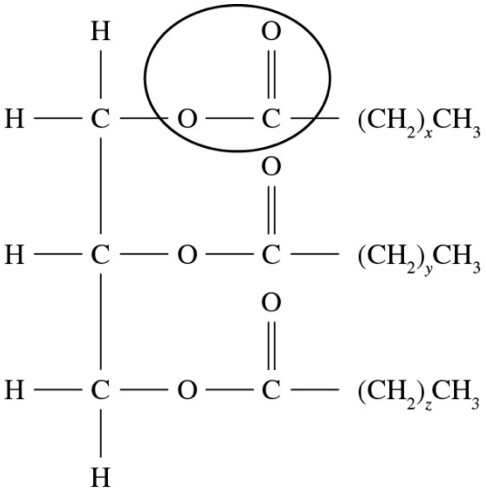
$2H^+(aq) + 2e^- \rightarrow H_2(g)$ 1 mark

Question 11c.

The only reactant, feedstock, needed for artificial photosynthesis is water. This means that the feedstock is renewable. 1 mark

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Question 11d.



2 marks

(1 mark for glycerol backbone, 1 mark for 3 × fatty acids connected correctly)

Question 11e.

–OCO– circled in **11d**; the major functional group in lipids is an ester. 1 mark

Question 11f.

A saturated fat will have only single C–C bonds, allowing it to stack more tightly, increasing the dispersion forces and the melting point. 1 mark

An unsaturated fat will have one or more C=C bonds, reducing the ability to stack closely, which reduces the dispersion forces and the melting point. 1 mark

Question 11g.

$n(I_2) = \frac{V}{V_m} = \frac{3.10}{24.8}$ = 0.125 mol

n(fatty acid) = $\frac{m}{M} = \frac{17.50}{280.0}$ = 0.0625 mol 1 mark

$n(I_2)$: n(fatty acid) = 1 : 2, degree of unsaturation = number of C=C bonds = 2 1 mark

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